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import pandas as pd

# Our Dataset
data = {
    'Humidity': ['High', 'High', 'Normal', 'High', 'Normal'],
    'Wind': ['Weak', 'Strong', 'Weak', 'Weak', 'Strong'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes']
}
df = pd.DataFrame(data)

def train_nb(df):
    # 1. Calculate Priors (How often does Play=Yes vs No happen?)
    priors = df['Play'].value_counts(normalize=True).to_dict()

    # 2. Calculate Likelihoods (e.g., If we played, how often was it Windy?)
    likelihoods = {}
    for feature in ['Humidity', 'Wind']:
        likelihoods[feature] = {}
        for outcome in ['Yes', 'No']:
            subset = df[df['Play'] == outcome]
            # Count feature values within each outcome
            likelihoods[feature][outcome] = subset[feature].value_counts(normalize=True).to_dict()
    return priors, likelihoods

def predict(new_case, priors, likelihoods):
    scores = {}
    for outcome in ['Yes', 'No']:
        # Start with the general probability (Prior)
        score = priors[outcome]
        # Multiply by conditional probability for each feature
        for feat, val in new_case.items():
            score *= likelihoods[feat][outcome].get(val, 0)
        scores[outcome] = round(score, 4)
    return scores

# Execution
priors, likelihoods = train_nb(df)
new_weather = {'Humidity': 'High', 'Wind': 'Weak'}
result = predict(new_weather, priors, likelihoods)

print(f"Final Prediction Scores: {result}")
# Output: {'Yes': 0.1333, 'No': 0.2} -> Predicts 'No'

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